ENGINEERING ETHICS CASE STUDY ANALYSIS

Case Study: "Quality of Product"

An in-depth examination of ethics, professional responsibility, personal perspectives, and business decisions in engineering.

Case Date: January 1, 1961 | ABC Company

Case Facts & Background

High Quality Legacy

ABC Company manufactured a product with a strong reputation for excellence across the industry and public for many years.

Market Competition

Competitors introduced lower-quality, lower-cost alternatives, causing a sharp decline in ABC Company's product sales.

Redesign Mandate

Management instructed engineers to redesign the product to cut costs while retaining the same established brand name.

Engineers Concern

> "Engineers questioned if retaining the same brand name for a lesser-quality product violates the canons of Ethics by misleading the public regarding expected standards."

The Core Ethical Question

> **"Do the engineers have a proper interest and ethical obligation to protest the company's decision, or to refuse to design a lower quality product?"**

Personal Opinion on the Dilemma

Risk of Deceit

Keeping the same product name without transparently communicating the quality reduction to the public can be deceitful.

Indirect Involvement

Assisting in the redesign as a professional engineer indirectly contributes to misleading consumers if transparency is lacking.

Direction of Fallout

Ultimately, unfavorable public reactions and brand fallout will target the company itself rather than individual engineers.

Public Safety vs. Commercial Scope

Public Safety (Canon 4)

The primary purpose of engineering canons is to protect public safety, health, and welfare. If a redesign compromises safety, Canon 4 clearly obligates the engineer to advise management immediately.

Individual Ethical Conduct

Canons and rules are restricted to individual ethical conduct. They do not govern general business decisions or commercial operational evaluations made by company management.

Management Responsibility & Communication



Management Determination

Offering a lower-quality product to meet market costs is exclusively a business decision. Management evaluates commercial risks and public acceptance.



Honest Professional Advice

Engineers are responsible for giving honest advice. They can voice concerns internally if they feel the decision is unwise or may mislead consumers, without refusing the work.

Conclusion: Professional Boundary

- **Internal Feedback:** Engineers should use internal channels to voice concerns regarding brand reputation or potential public misunderstanding.
- **Corporate Backlash:** If the public is misled by quality drops, the unfavorable reaction will fall upon the company, not the engineer.
- **Preserving Engineering Scope:** Attempting to extend engineering ethics canons to interfere in commercial business decisions would do a disservice to the profession.